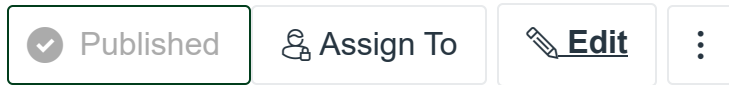


Digitizing



Digitizing UF Campus

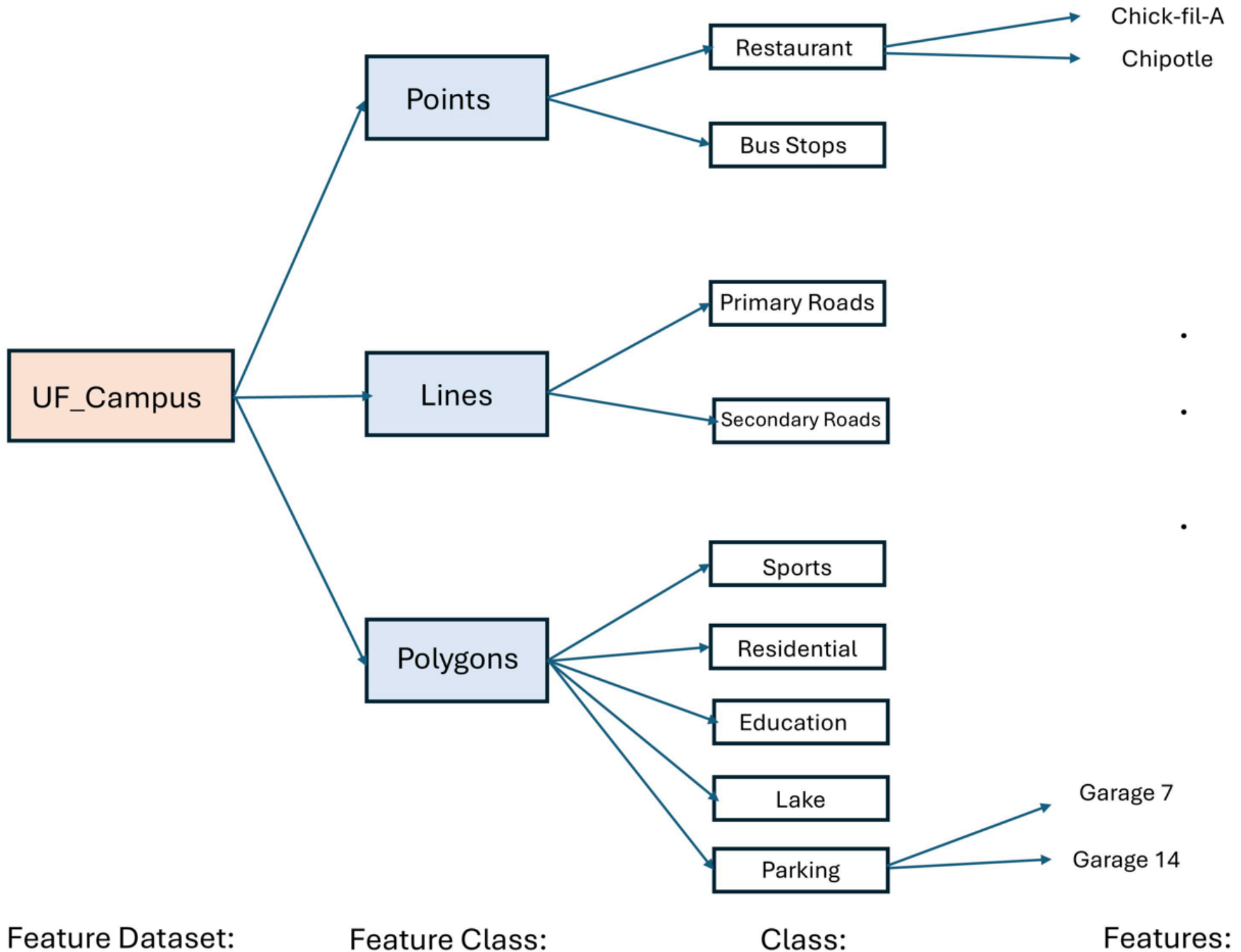
Objective:

Digitizing lets you make personal maps, such as tracking national parks you've visited or highlighting your favorite places, bringing your own experiences to life on a map.

In this lab, we build a geospatial data model for UF campus using ArcGIS Pro.

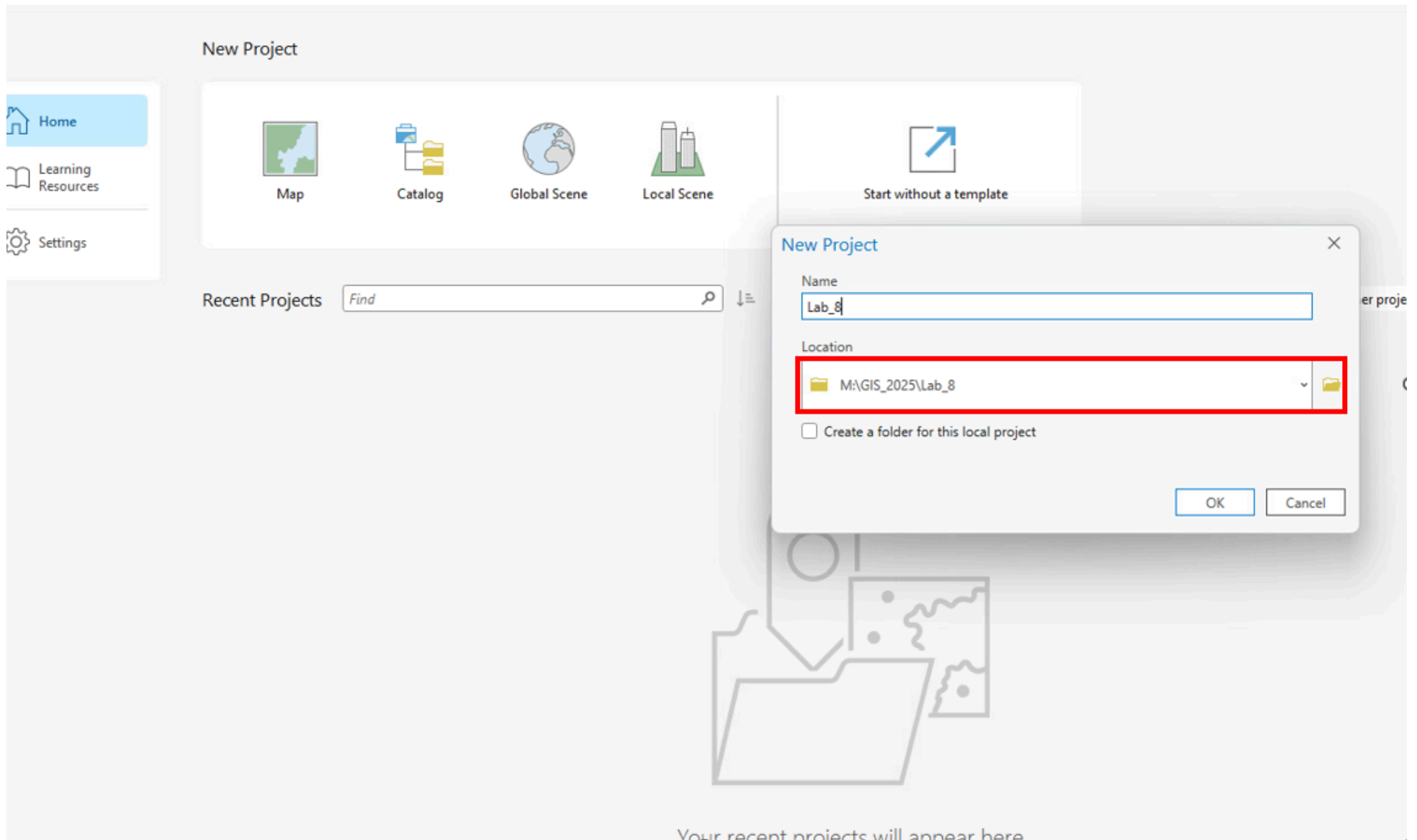
- Create a Feature Dataset with point, line, and polygon Feature Classes.
- Represent real-world campus objects such as buildings, lakes, roads, restaurants, and bus stops in these layers.
- Assign descriptive attributes to each feature for effective organization and analysis.
- Learn best practices in GIS data creation, organization, and attribute management.

To create the geodatabase of the UF campus, we will organize it into three separate Feature Classes: Points, Lines, and Polygons. Within each feature class, we will define the following unique Classes:

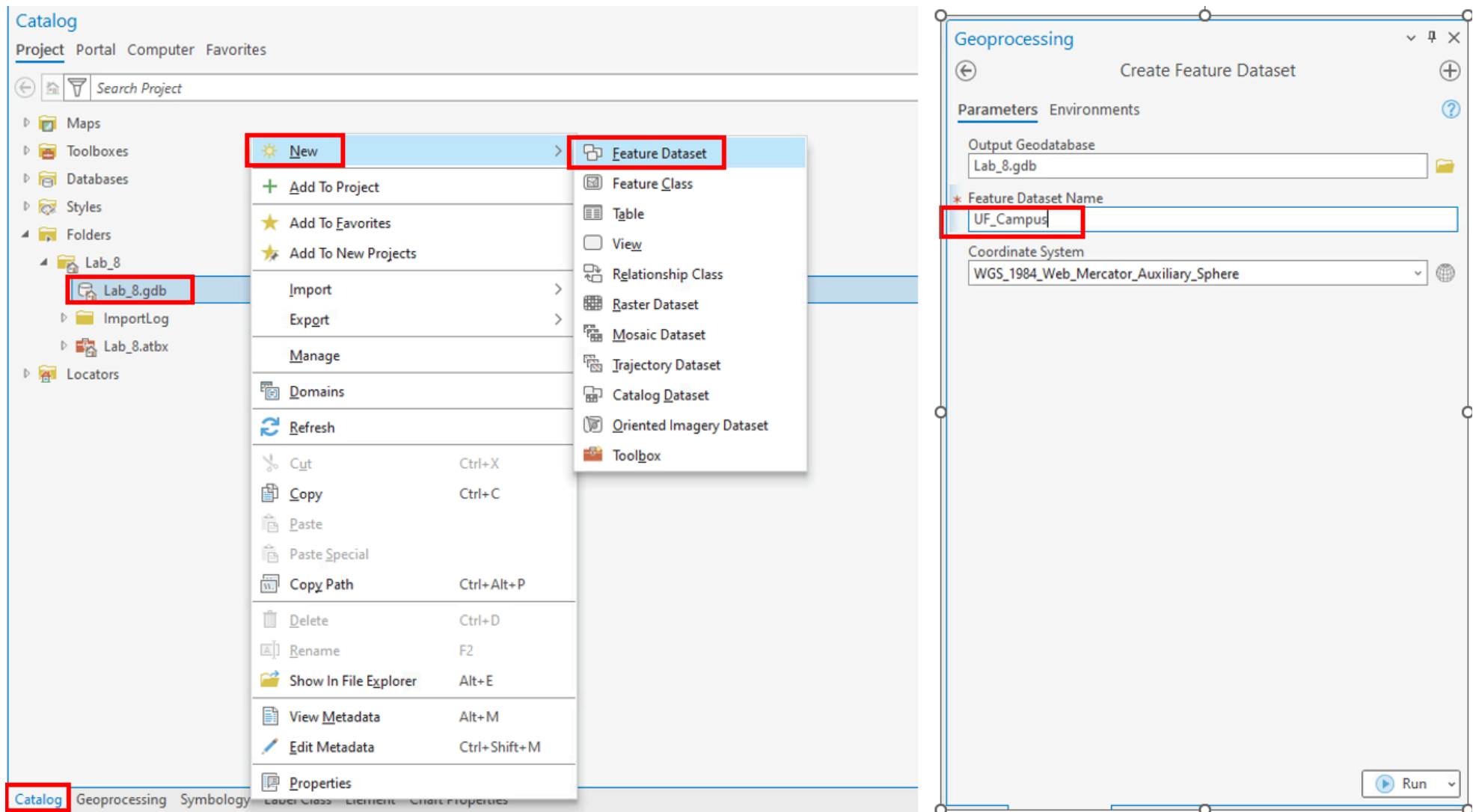


Steps:

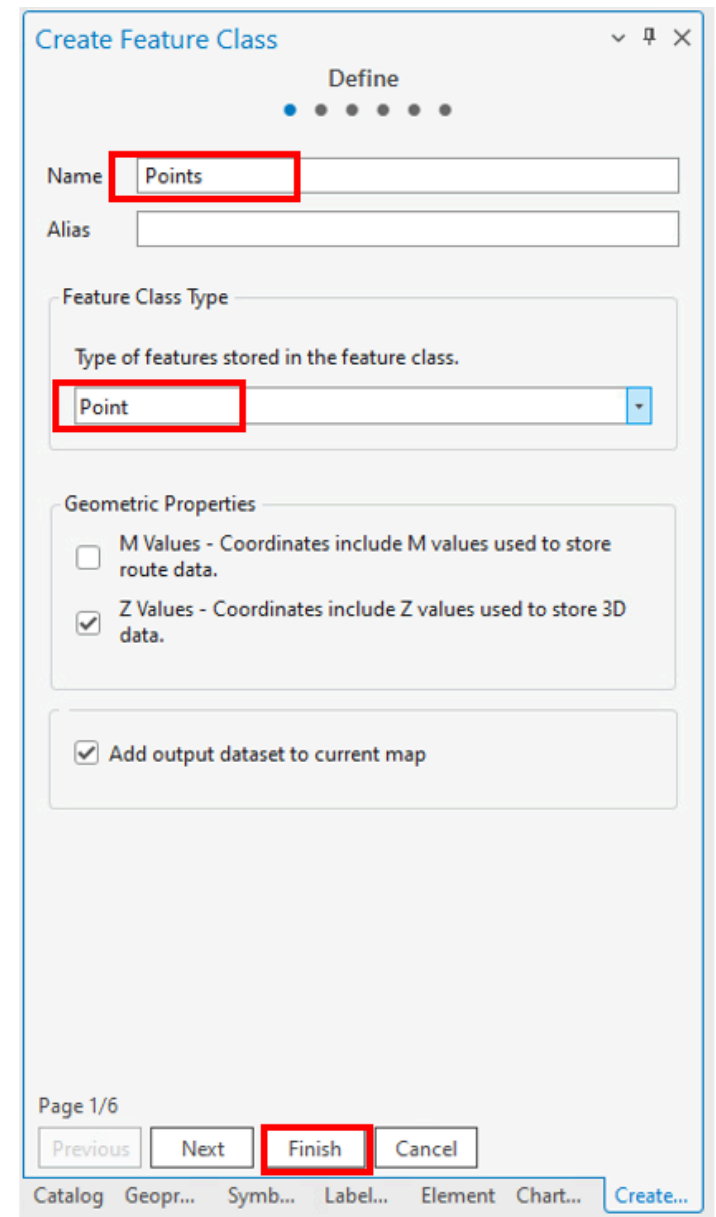
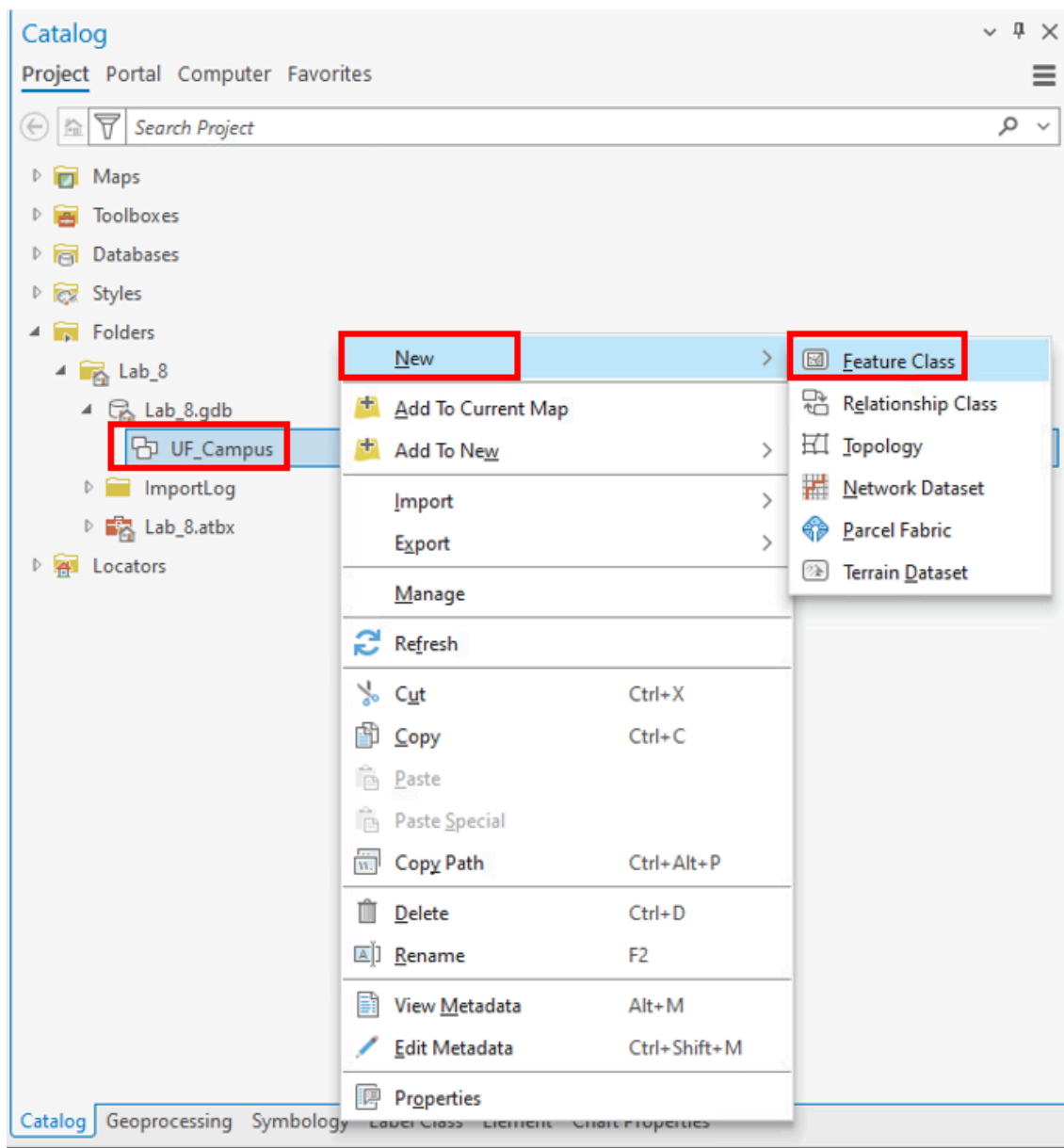
1. Open ArcGIS Pro, create a new map project, and save the project in the lab folder.



2. In the Catalog pane, right-click your project geodatabase (**Lab_8.gdb**), select **New > Feature Dataset**, enter **UF_Campus** as the name, and then click **Run** to create the dataset.

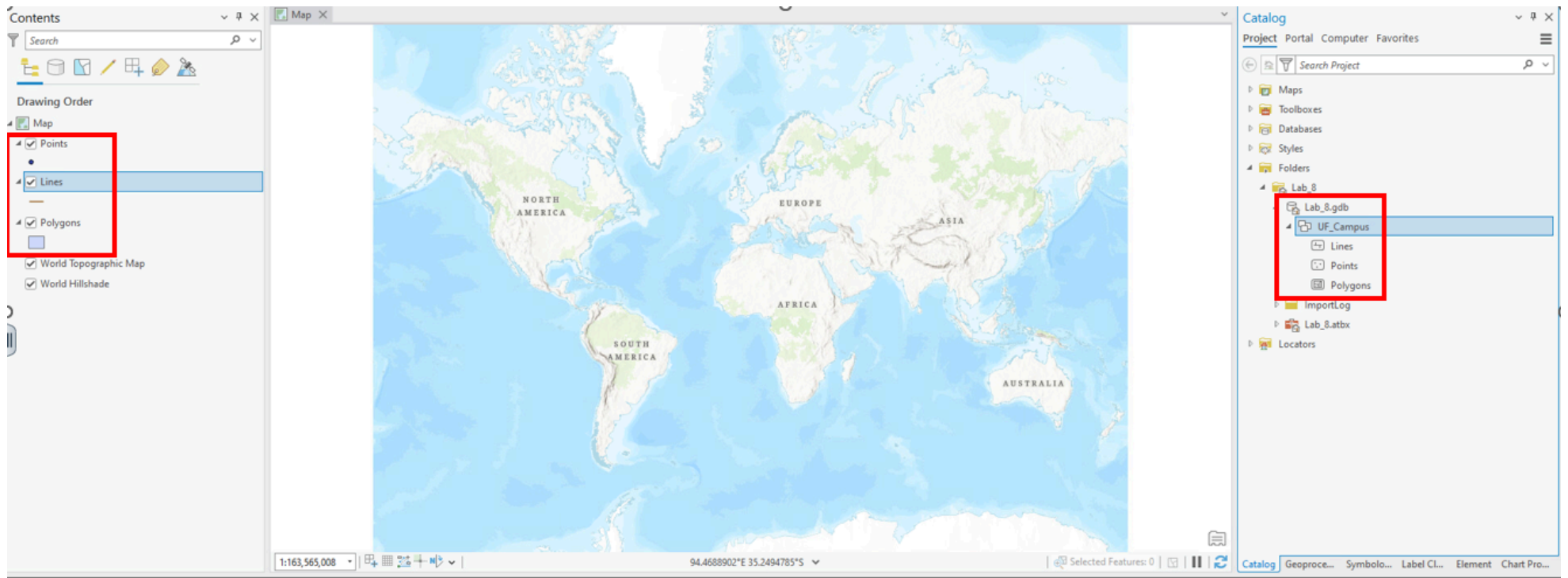


3. Right-click **UF_Campus**, select **New > Feature Class**, name it **Points**, choose **Point** as the feature type, and click **Finish** to create the feature class.

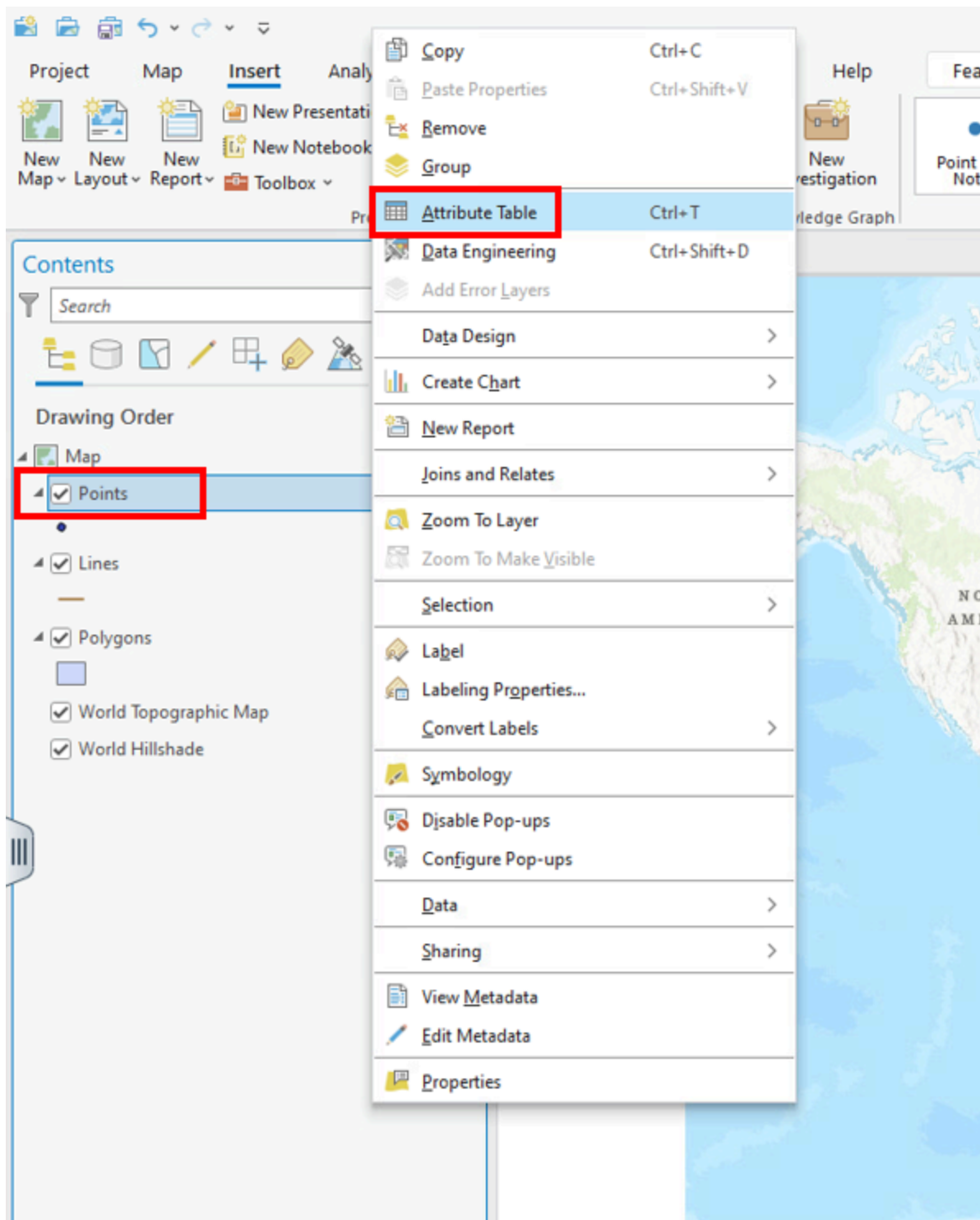


4. Repeat the previous step to create two more feature classes within **UF_Campus**: name one **Polygons** and select **Polygon** as the type; name the other **Lines** and select **Line** as the type.

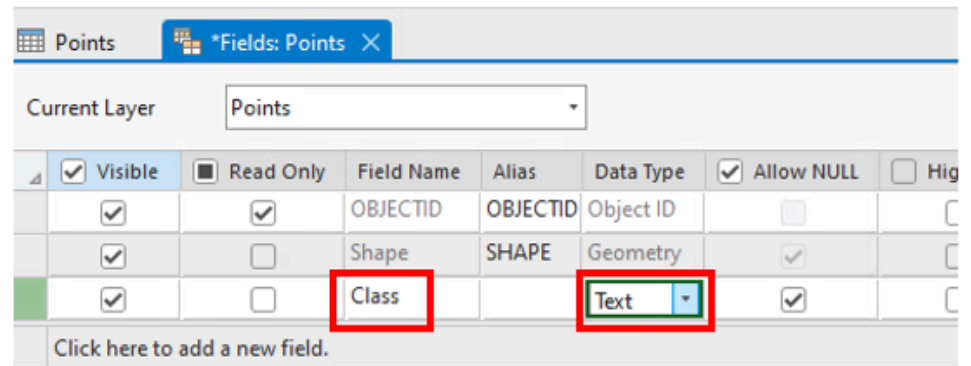
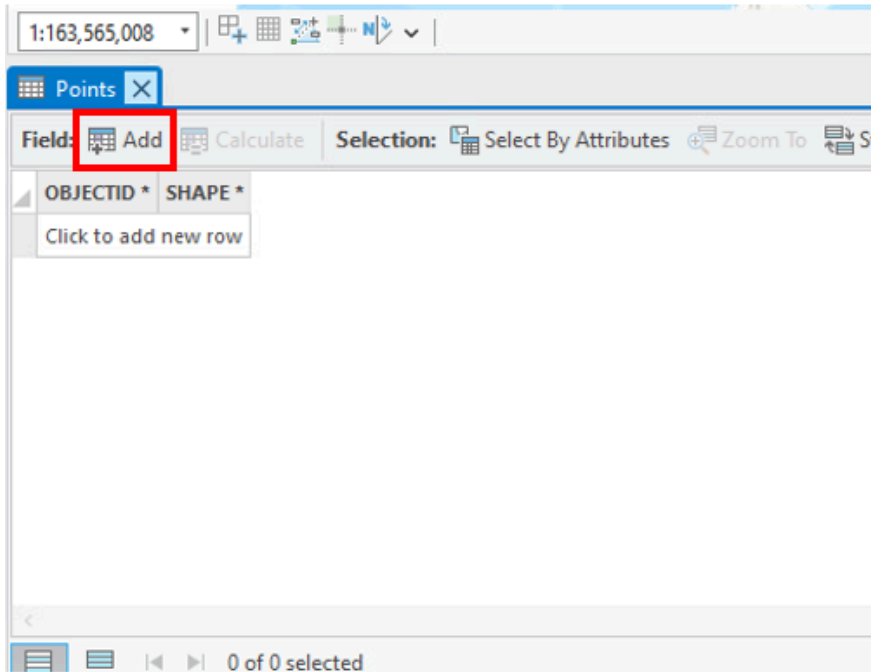
After this, you should see all three empty feature classes (Points, Lines, and Polygons) in the Contents pane.



5. In the Contents pane, right-click the **Points** feature class and select **Attribute Table** to open it.



6. Click **Add** to add a new field (column), enter **Class** as the field name, and choose **Text** as the data type.



7. Add another field, enter **Name** as the field name, select **Text** as the data type, and click **Save**.

Lab_8 Command Search (Alt+Q)

Project Fields Insert Analysis View Share Help

Clipboard Data Design Manage Edits Edits Filter Field Order Data Engineering

Contents

Search

Drawing Order

- Map
- ☒ Points
- ☒ Lines
- ☒ Polygons
- ☒ World Topographic Map
- ☒ World Hillshade

Map

1:163,565,008 152.4016982°E 50.5083092°N

Points *Fields: Points

Current Layer Points

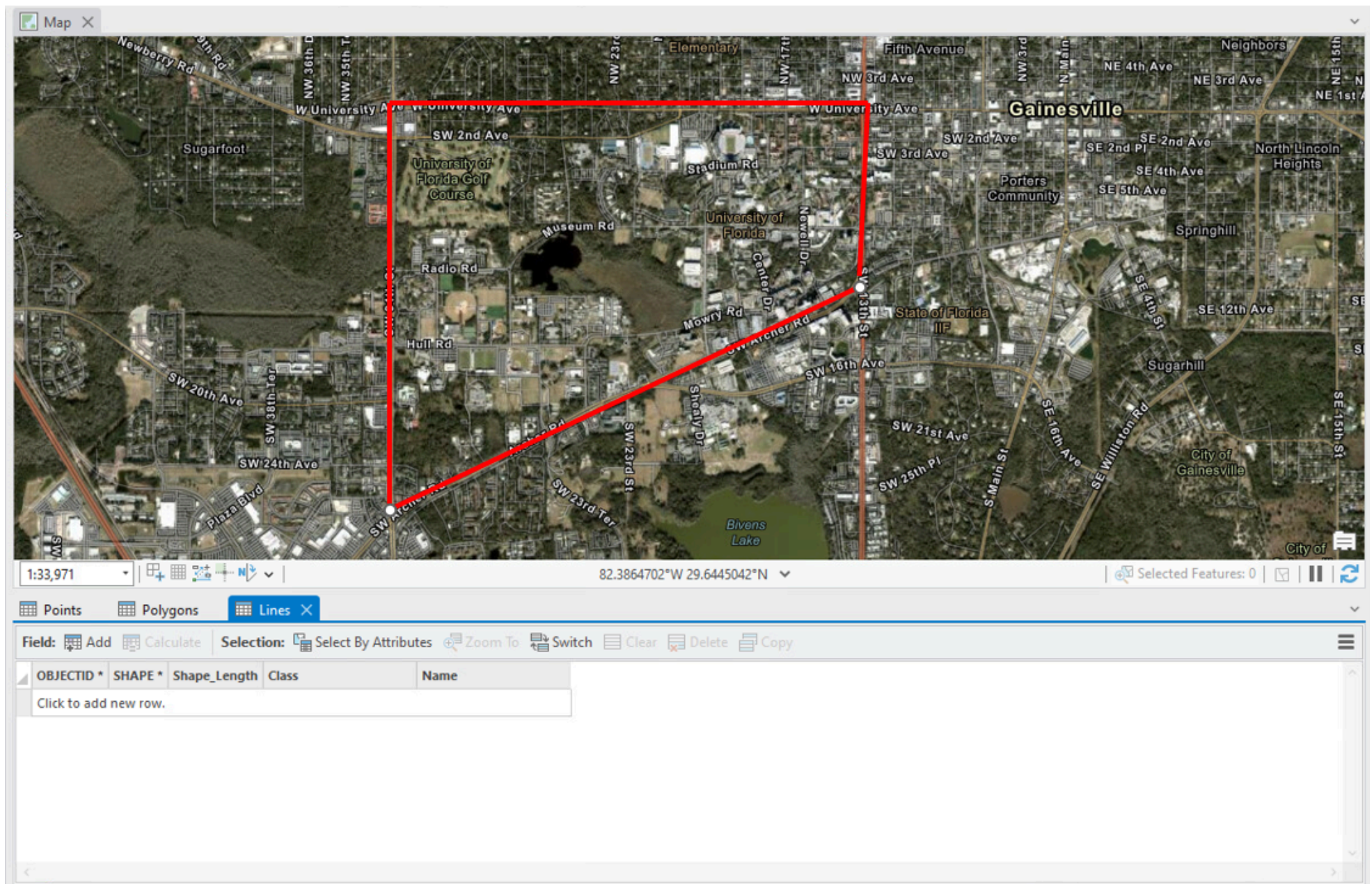
	Visible	Read Only	Field Name	Alias	Data Type	Allow NULL	Highlight	Number Format	Domain	Default	Length
	☒	☒	OBJECTID	OBJECTID	Object ID	☐	☐	Numeric			
	☒	☐	Shape	SHAPE	Geometry	☒	☐				
	☒	☐	Class		Text	☒	☐				255
	☒	☐	Name		Text	☒	☐				255

Click here to add a new field.

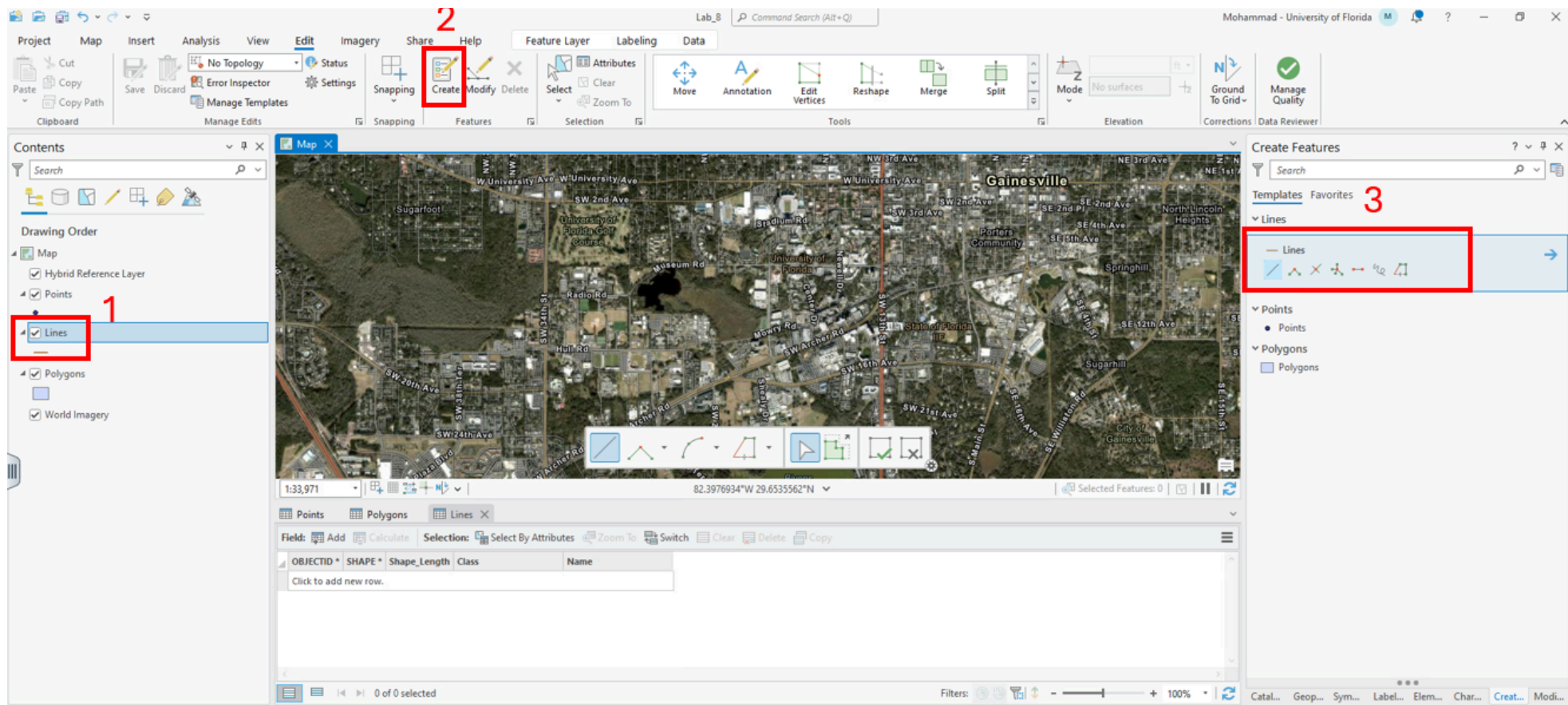
8. Repeat steps 5–7 to add the **Class** and **Name** fields to both the **Lines** and **Polygons** feature classes.

Now that the three feature classes and their fields are ready, you'll begin populating them by drawing features on the UF campus map. We will start with the **Lines** feature class: group roads into two classes (**Primary** and **Secondary**) and digitize 3 to 4 roads for each class.

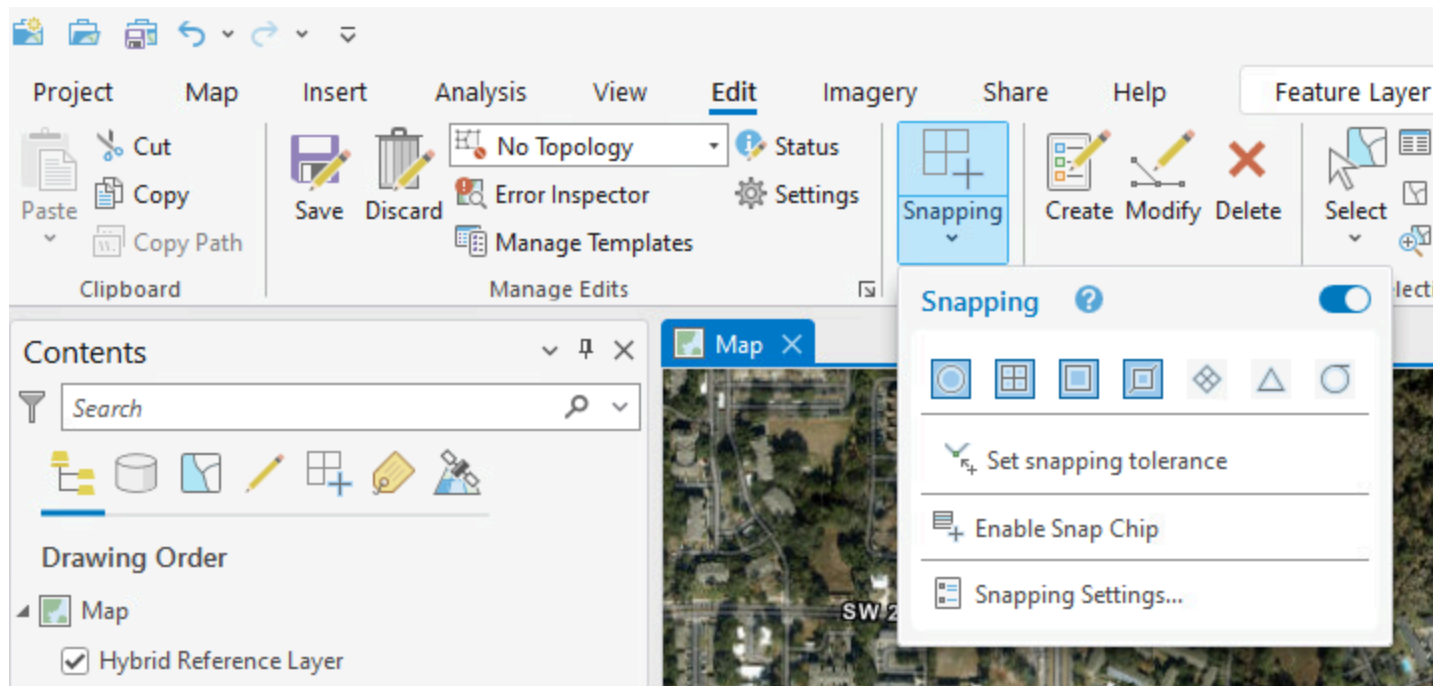
10. Change the basemap to **Imagery Hybrid** for better visual reference. Then, open the attribute table for the **Lines** feature class. You will digitize several features within the highlighted area.



11. In the Contents pane, select **Lines**. Go to the **Edit** tab, click **Create**, and in the Create Features window, choose **Lines** to begin digitizing.



12. In the **Edit** tab, activate the **Snapping** tool to help align your features accurately.



13. To digitize the first primary road, **Archer Road**, left-click to set the starting point, continue left-clicking to add segments along the road, and double-click at the endpoint to finish the line.

Map X

1:53,080 82.4029921°W 29.6461313°N

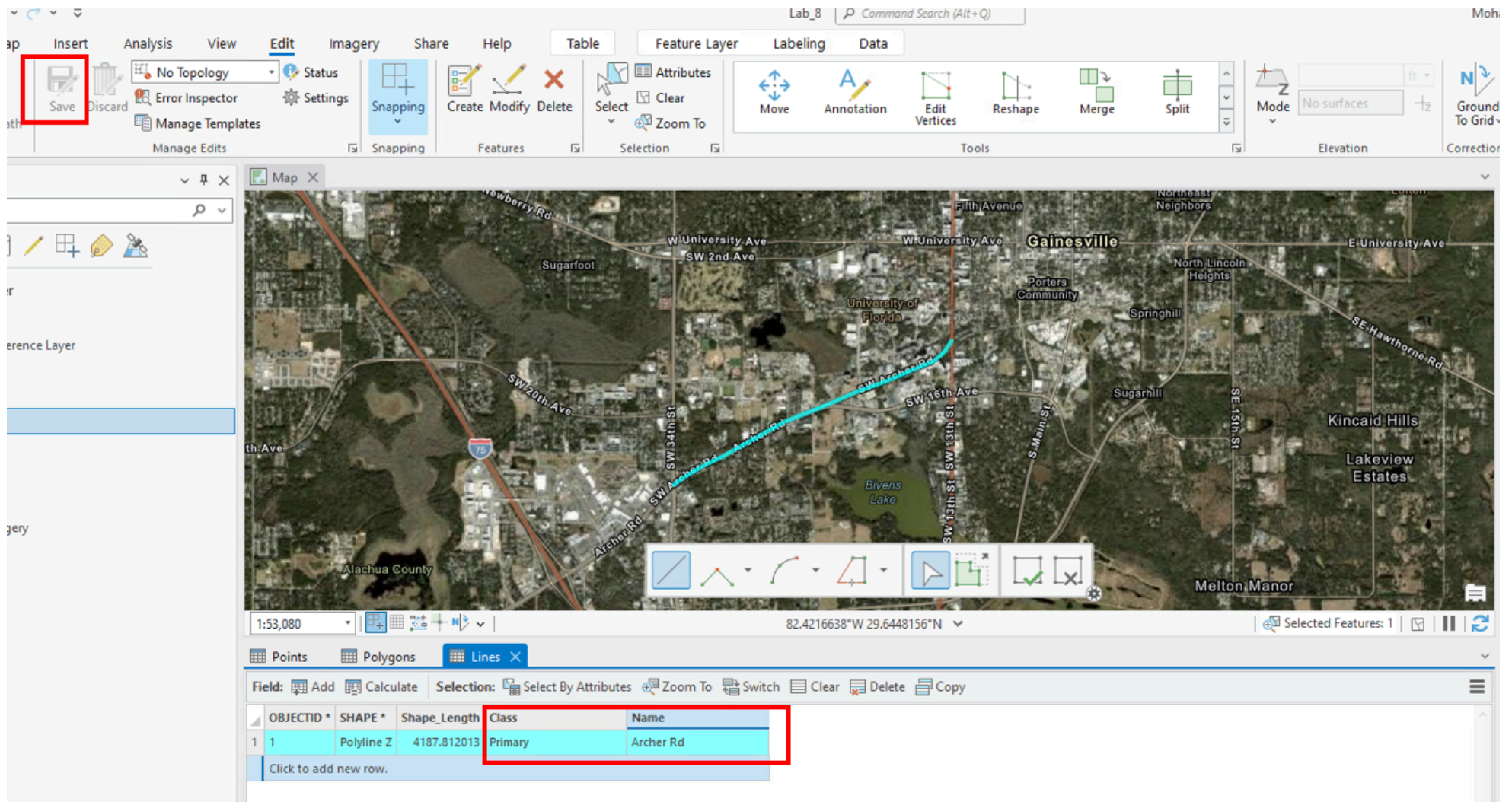
Points Polygons Lines X

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

OBJECTID *	SHAPE *	Shape_Length	Class	Name
1	Polyline Z	4187.812013	<Null>	<Null>

Click to add new row.

14. In the attribute table, enter **Primary** for the Class field and **Archer Rd** for the Name field, then click **Save** on the ribbon .



15. Adjust the symbology for the **Lines** feature class by right-clicking it in the Contents pane, selecting **Symbology**, and changing the color and line width for better visual representation.
16. Digitize three more primary roads and three secondary roads, filling in the attribute table with the appropriate class and name for each feature (use Google Maps to help identify road locations and names, and remember to save your edits regularly).

The screenshot displays the ArcGIS Desktop interface. The **Edit** tab is active, showing various editing tools. The **Contents** pane on the left shows the **Lines** feature class selected. The main map area shows a digitized road network in Gainesville, Florida. Below the map, a table lists the digitized road features.

OBJECTID *	SHAPE *	Shape_Length	Class	Name
1	Polyline Z	4187.812013	Primary	Archer Rd
2	Polyline Z	1306.287579	Primary	SW 13th St
3	Polyline Z	3687.244232	Primary	W University Ave
4	Polyline Z	3221.195397	Primary	SW 34th St
5	Polyline Z	2338.275803	Secondary	SW 2nd Ave
6	Polyline Z	3495.830632	Secondary	Museum Dr
7	Polyline Z	2030.457036	Secondary	Hull Rd

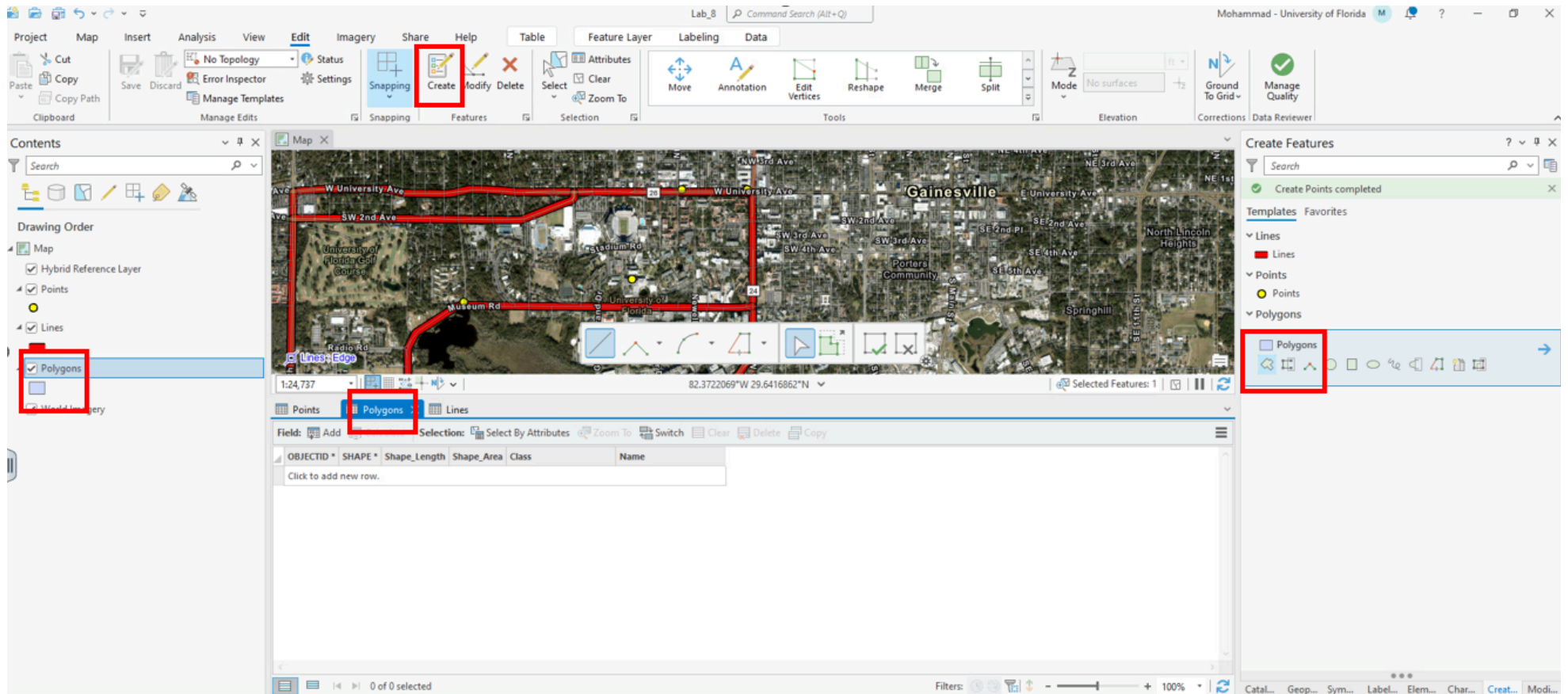
17. To digitize Points, first adjust their size and color for better visibility in the **Symbology**. Then, select the **Points** feature class in the **Contents** pane, go to **Create** in the **Edit** tab, and choose **Points** in **Create Features** window.

Add four points for the **Restaurant** class by clicking their locations on the map, entering **Class** and **Name** in the **Points** attribute table. Next, add four points for the **Bus Stop** class (leave the Name field blank for these), and remember to save your edits.

The screenshot shows the ArcGIS Desktop interface. The **Edit** tab is active, and the **Save** button is highlighted with a red box. The **Contents** pane on the left shows the **Points** layer selected, also highlighted with a red box. The main map area displays a street map of Gainesville, Florida, with a red polygon outline and several yellow points. The attribute table at the bottom is also highlighted with a red box and contains the following data:

OBJECTID *	SHAPE	Class	Name
1	Point Z	Restaurant	Chick-fil-A
2	Point Z	Restaurant	Raising Cane's
3	Point Z	Restaurant	Los Pollos
4	Point Z	Restaurant	Subway
5	Point Z	Bus Stop	<Null>
6	Point Z	Bus Stop	<Null>
7	Point Z	Bus Stop	<Null>
8	Point Z	Bus Stop	<Null>

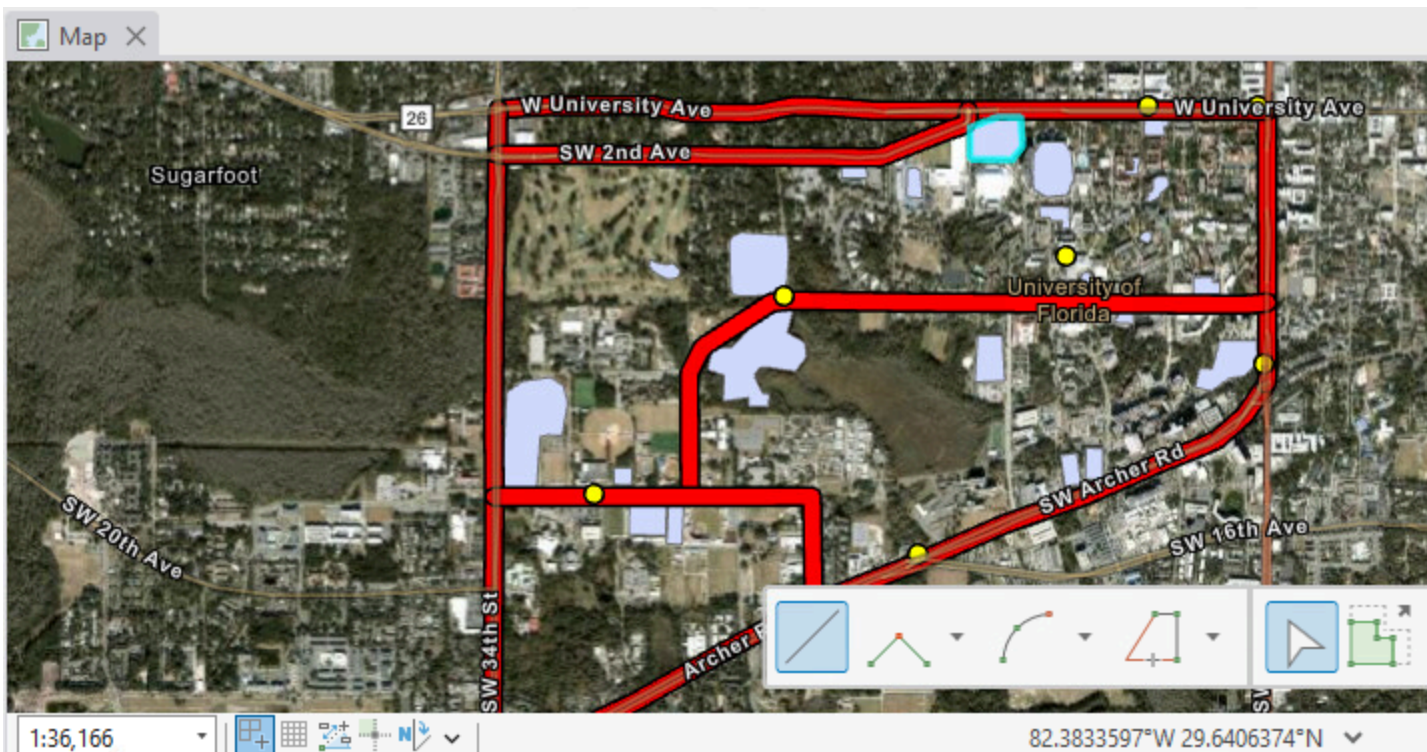
17. To create polygon features, select **Polygons** in the Contents pane, go to the **Edit** tab, click **Create**, and choose **Polygons** in the Create Features window.



18. For Polygons, use these five **Classes**:

- **Sports** (gyms, football fields, soccer fields, baseball fields, etc.)
- **Residential** (student housings)
- **Education** (departments)
- **Lake** (lakes and ponds)
- **Parking**

Add at least four polygon features for each Class by drawing them on the map. For each polygon, enter the **Class** name in the attribute table. You do not need to enter names (but you may if you wish). Remember to save your edits.



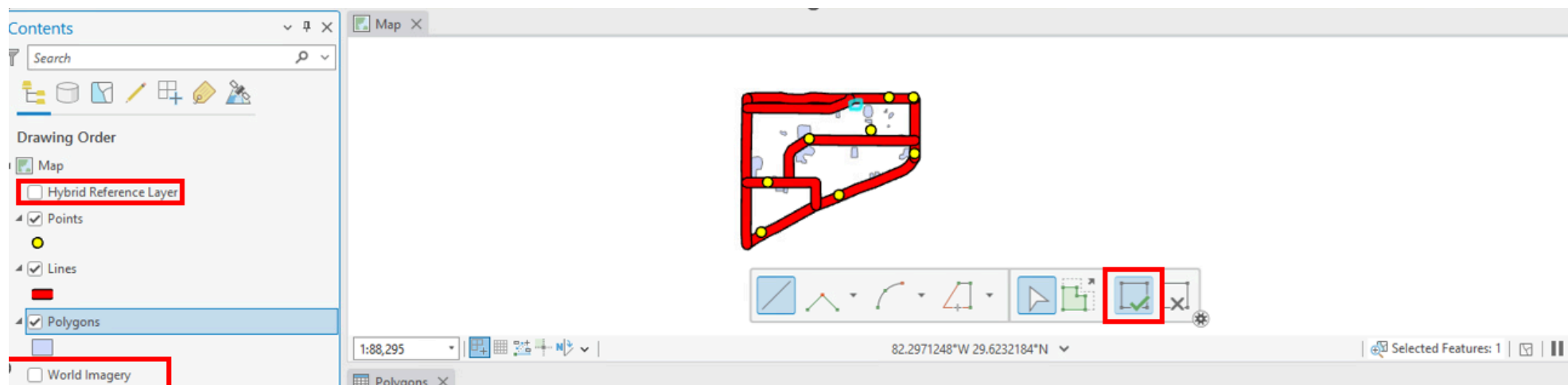
Polygons

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

	OBJECTID *	SHAPE *	Shape_Length	Shape_Area	Class	Name
1	1	Polygon Z	926.54572	43127.449729	Residential	< Null >
2	2	Polygon Z	507.514659	15699.105961	Residential	< Null >
3	3	Polygon Z	1078.652841	79878.788963	Residential	< Null >
4	4	Polygon Z	1523.354351	116661.273263	Residential	< Null >
5	5	Polygon Z	327.755307	2870.174565	Residential	< Null >
6	6	Polygon Z	1816.263305	113885.501601	Lake	< Null >
7	7	Polygon Z	389.969621	8324.119154	Lake	< Null >
8	8	Polygon Z	200.844126	3038.992994	Lake	< Null >
9	9	Polygon Z	178.460857	2270.789681	Lake	< Null >
10	10	Polygon Z	370.537417	7083.658452	Education	< Null >
11	11	Polygon Z	391.418664	7844.148964	Education	< Null >
12	12	Polvaon Z	305.486741	5296.502022	Education	< Null >

13	13	Polygon Z	498.357169	10312.377411	Education	<Null>
14	15	Polygon Z	332.747467	6914.722094	Sports	<Null>
15	16	Polygon Z	788.026681	44576.758206	Sports	<Null>
16	17	Polygon Z	412.264981	9733.179955	Sports	<Null>
17	18	Polygon Z	411.302098	10250.036388	Sports	<Null>
18	19	Polygon Z	820.886276	45796.354413	Parking	<Null>
19	20	Polygon Z	704.101615	28931.112259	Parking	<Null>
20	21	Polygon Z	525.563789	13911.90847	Parking	<Null>
21	22	Polygon Z	463.743743	12297.094004	Parking	<Null>
22	23	Polygon Z	458.946797	11896.952416	Parking	<Null>
Click to add new row.						

19. Unselect the basemap layer, then click the **Finish Editing** button to complete your edits.



20. Open the **Symbology** for each layer, choose **Unique Values**, and set the field to **Class**. Click **Add All Values** if all classes do not appear. Adjust the size and color for each class to create a clear and visually effective map.

Symbology - Points



Primary symbology

Unique Values

Field 1 Class

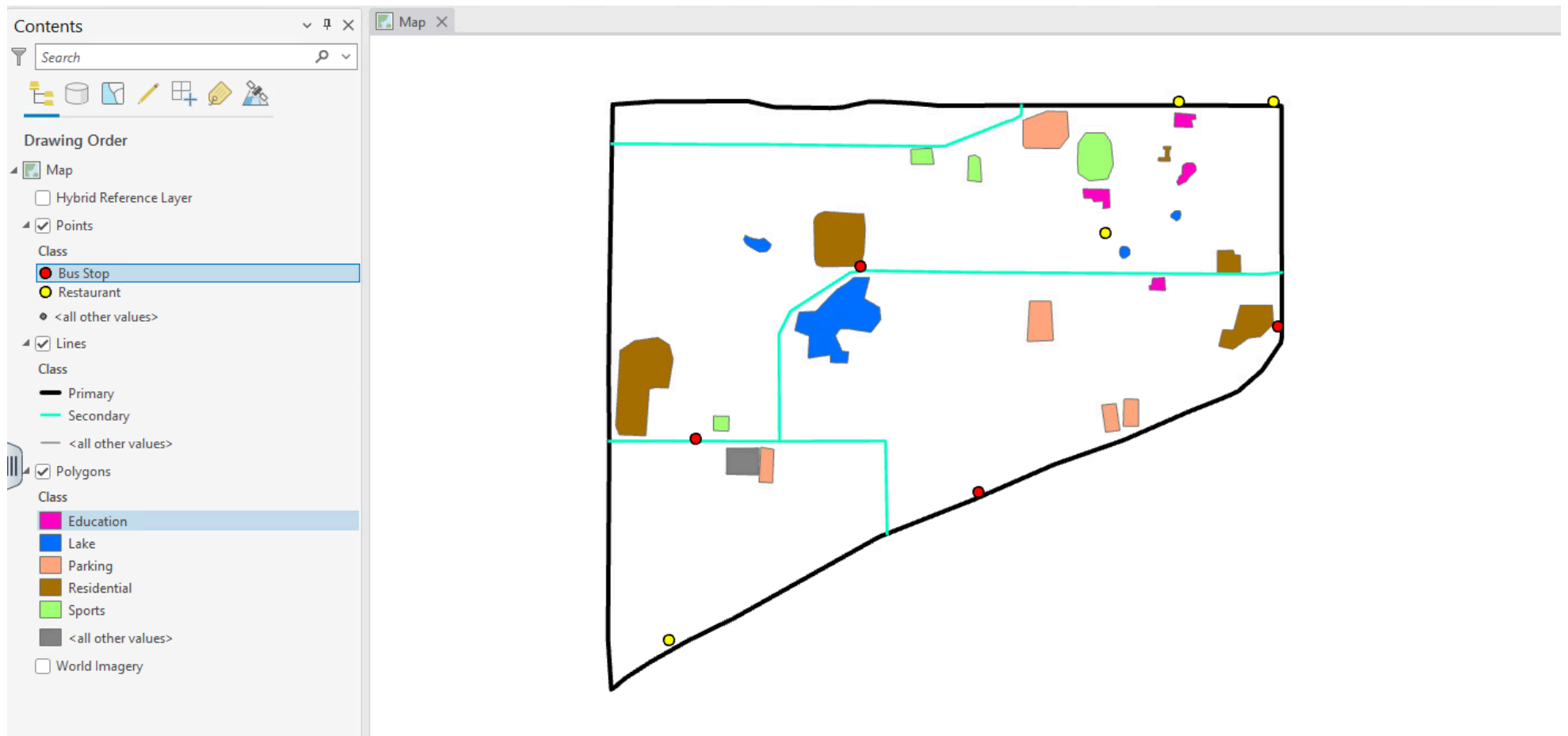
Add field

Color scheme

Classes Scales



Symbol	Value	Label	
▼ Class 2 symbol classes ...			
•	Bus Stop	Bus Stop	
•	Restaurant	Restaurant	
▼ <all other values>			
•	<all other value...>	<all other value...>	



21. Add labels for Points and Lines by right-clicking each layer and selecting **Label**. Then, right-click again and open Labeling Properties and set the label field to **Name**.

Label Class - Points

Class 1

Class  Symbol Position



Language  Arcade

Title  Custom

Fields



Functions



OBJECTID

Class

Name

Abs()

Acos()

Angle()

Any()

 Insert Values

Expression

`$feature.Name`



☒ Insert fields with coded value description function

☒ Remove extra spaces

☐ Remove extra line breaks

[Learn more about label expressions](#)

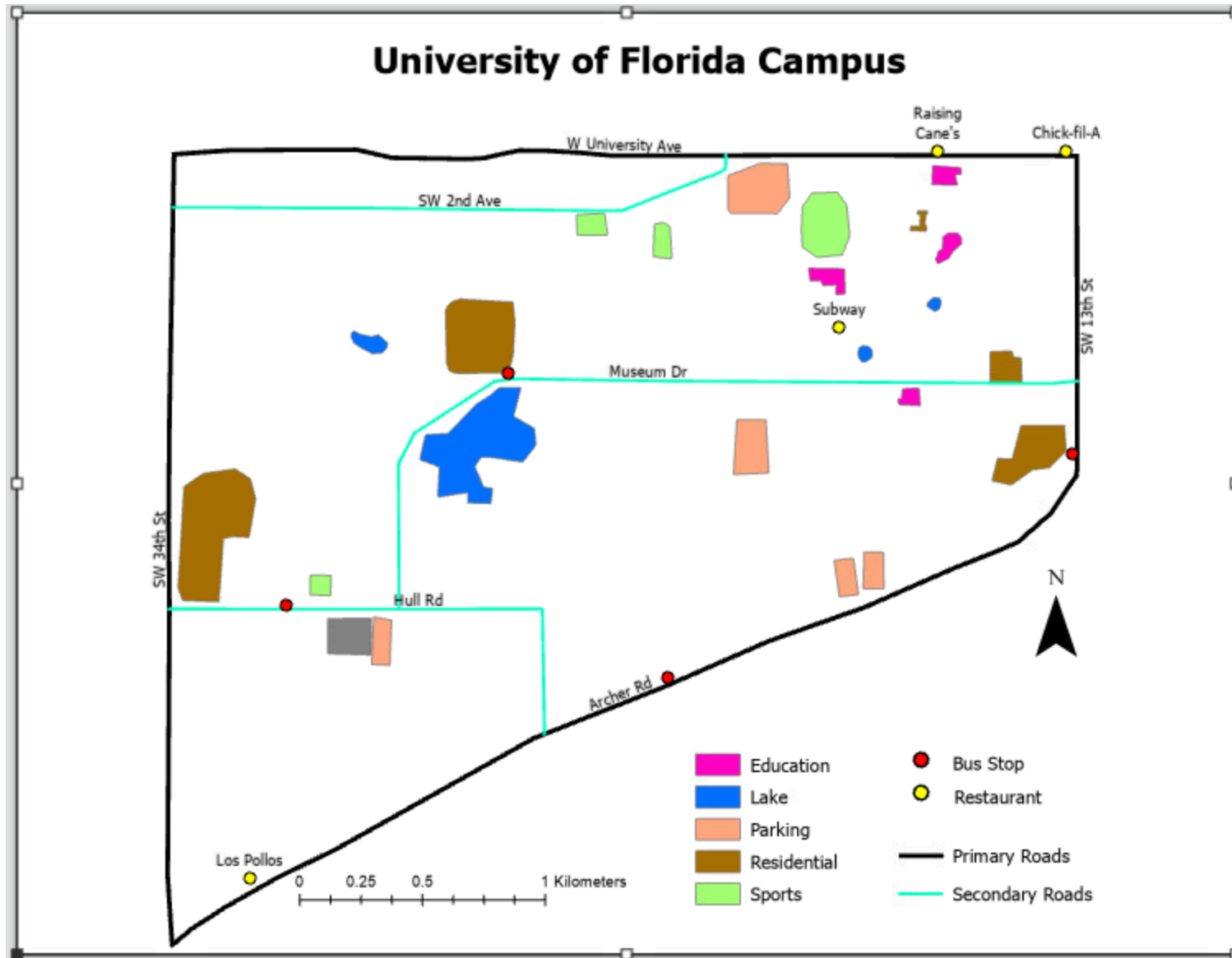
Apply

Cancel

Catalog Geoprocessi... Symbology **Label Class** Element Chart Propert... Modify Featu...

22. Make a Map

- From the Insert tab, create a new layout and add your map frame.
- Include a title, legend, scale bar, north arrow, and your identification (name, date, etc.) on the map.
- Export your finished map as a PDF and submit it as your assignment.



Remember, this is not just drawing shapes: every feature you create is tied to real spatial coordinates and a database of attributes. This means your map can be measured for true distances and areas, and it supports spatial queries.

For example, if you want to answer a question like "Find all student housing on campus that are within one mile of a gym, have parking within 500 feet, have at least four restaurants nearby, and are not too close to primary roads, etc." a GIS dataset like this would allow you to do so if we continued creating all features accurately for the entire campus. We will learn more about spatial queries in future labs.

Points 10

Submitting a file upload

Due	For	Available from	Until
Sep 28	Everyone	Sep 22 at 12am	Oct 3 at 11:59pm

+ Rubric